1) List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

IPv4 vs IPv6 Comparison

- IPv4 uses 32-bit addresses, while IPv6 uses 128-bit addresses.
- IPv4 is commonly written in dotted-decimal form, such as 192.168.10.1. IPv6 is commonly written in hexadecimal groups separated by colons, such as 2001:db8::1.
- IPv4 provides a much smaller address space, while IPv6 provides a vastly larger address space.
- IPv4 commonly uses subnet masks such as 255.255.255.0 or CIDR notation such as /24. IPv6 uses prefix lengths such as /64.
- IPv4 networks commonly use private address ranges such as 192.168.0.0/16. IPv6 has different address types and does not use IPv4-style private addressing in the same way.
- IPv6 was designed to provide a long-term solution to IPv4 address exhaustion.

Key Difference Table

- IPv4 — 32-bit address — Example: 192.168.10.1 — Smaller address space.
- IPv6 — 128-bit address — Example: 2001:db8::1 — Much larger address space.

2) Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Step 1: Choose the Network

- Network: 192.168.10.0/24.
- Subnet mask: 255.255.255.0.
- Default gateway: 192.168.10.1.

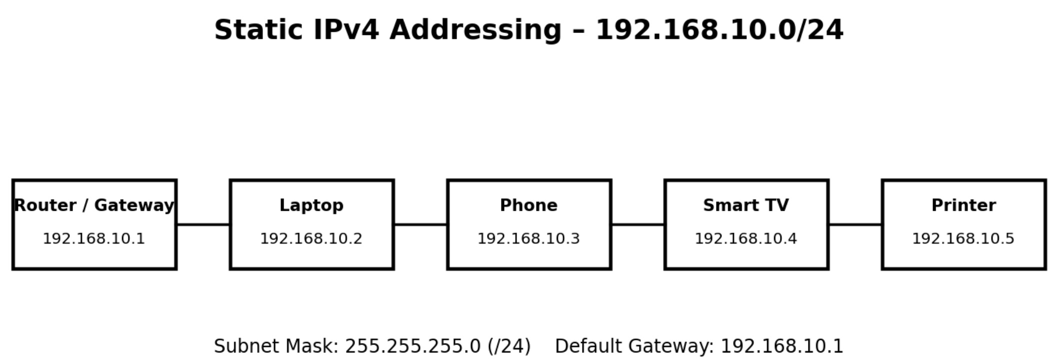
Step 2: Assign Static IP Addresses

- Laptop: 192.168.10.2 — Subnet Mask: 255.255.255.0 — Gateway: 192.168.10.1.
- Phone: 192.168.10.3 — Subnet Mask: 255.255.255.0 — Gateway: 192.168.10.1.
- Smart TV: 192.168.10.4 — Subnet Mask: 255.255.255.0 — Gateway: 192.168.10.1.
- Printer: 192.168.10.5 — Subnet Mask: 255.255.255.0 — Gateway: 192.168.10.1.

Addressing Table

- Laptop → 192.168.10.2 / 255.255.255.0 / 192.168.10.1
- Phone → 192.168.10.3 / 255.255.255.0 / 192.168.10.1
- Smart TV → 192.168.10.4 / 255.255.255.0 / 192.168.10.1
- Printer → 192.168.10.5 / 255.255.255.0 / 192.168.10.1

Reference Diagram



- The diagram shows the four devices using unique addresses in the 192.168.10.0/24 network.

3) Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

Step 1: Calculate the Number of Subnets

- Assume the original network is 192.168.1.0/24.
- Changing /24 to /27 borrows 3 bits for subnetting.
- Number of subnets = 2^3 = 8 valid subnets.

Step 2: Calculate Hosts per Subnet

- A /27 leaves 5 host bits.
- Total addresses per subnet = 2^5 = 32.
- Usable hosts per subnet = 32 − 2 = 30.

Step 3: Find the Block Size

- The /27 subnet mask is 255.255.255.224.
- Block size = 256 − 224 = 32.

First Three Subnets

- Subnet 1: Network 192.168.1.0 — Broadcast 192.168.1.31 — Usable hosts 192.168.1.1–192.168.1.30.
- Subnet 2: Network 192.168.1.32 — Broadcast 192.168.1.63 — Usable hosts 192.168.1.33–192.168.1.62.
- Subnet 3: Network 192.168.1.64 — Broadcast 192.168.1.95 — Usable hosts 192.168.1.65–192.168.1.94.

Final Answer

- Valid subnets: 8.
- Usable hosts per subnet: 30.
- Subnet mask: 255.255.255.224 (/27).

4) You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Step 1: Convert the Subnet Mask to CIDR

- Subnet mask: 255.255.255.0.
- This is equivalent to /24.

Step 2: Find the Network Address

- IP address: 10.0.5.17.
- Subnet mask: 255.255.255.0.
- Using a bitwise AND, the network address is 10.0.5.0.

Step 3: Find the Broadcast Address

- The /24 network covers the range from 10.0.5.0 to 10.0.5.255.
- The broadcast address is therefore 10.0.5.255.

Step 4: Find the Valid Host Range

- Network address 10.0.5.0 cannot be assigned to a normal host.
- Broadcast address 10.0.5.255 cannot be assigned to a normal host.
- Valid host addresses are 10.0.5.1 through 10.0.5.254.

Final Answer

- Network: 10.0.5.0/24.
- Broadcast: 10.0.5.255.
- Valid host range: 10.0.5.1–10.0.5.254.

5) Use an online subnet calculator tool to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

Step 1: Open a Subnet Calculator

- Open a trusted online subnet calculator.
- Enter IP address: 10.0.5.17.
- Enter subnet mask or CIDR: 255.255.255.0 or /24.

Step 2: Compare the Results

- Compare the calculator's network address, broadcast address, and valid host range with the manual calculation from Task 4.

Verification Result

- Manual answer: Network 10.0.5.0, Broadcast 10.0.5.255, Valid hosts 10.0.5.1–10.0.5.254.
- Expected verification statement: The manual answer matched the subnet calculator output.